

IoT Edge Stack

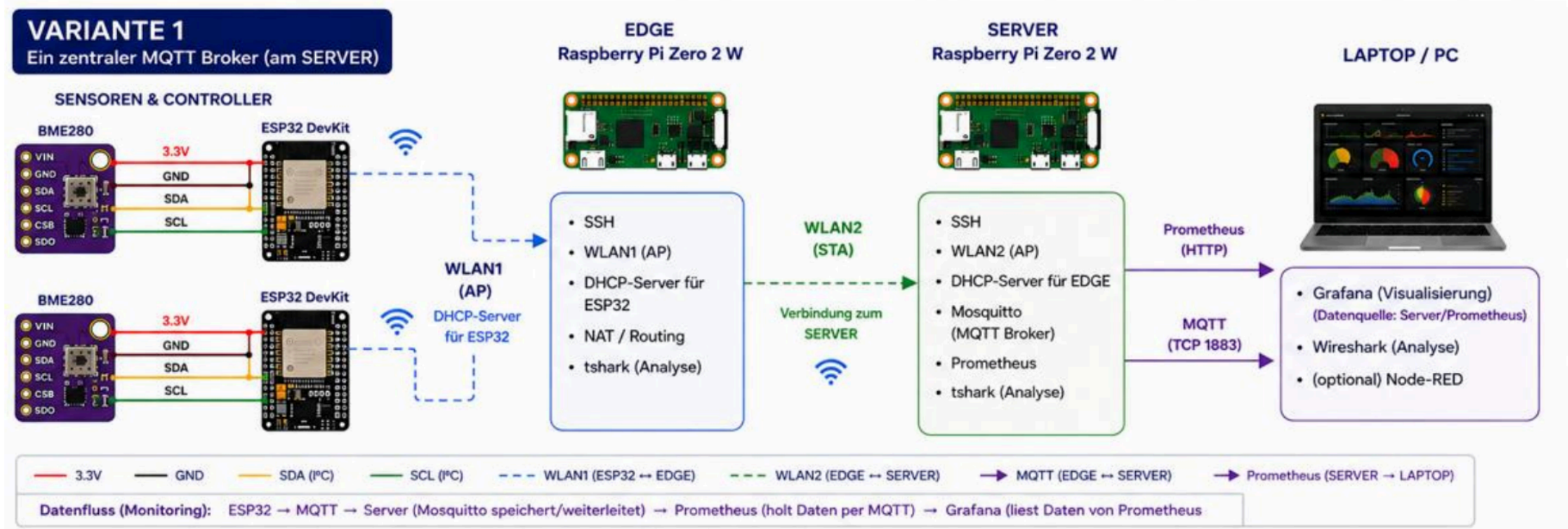
End-to-end sensor telemetry on Raspberry Pi

ESP32 · MQTT · Prometheus · Grafana · Docker · Ansible

The problem

- Collect environmental data (temperature, humidity, pressure) from distributed **ESP32 sensor nodes**.
- Keep the sensors on an **isolated network**, separate from the backbone.
- Deliver the data reliably to a **central monitoring backend**.
- Make the whole setup **reproducible and automatable**.

Architecture



Design decision: one radio per role

- The edge node must be an **access point** *and* an **upstream client** at once.
- A single on-board Wi-Fi chip allows this only on **one shared channel** and was **unreliable for concurrent data traffic**.
- **Solution:** a second (USB) Wi-Fi adapter — **AP on-board, uplink on USB**.
- Result: independent channels, normal DHCP, standard tooling, reliable.

Engineering trade-off documented as an architecture decision in the repo.

Observability pipeline

MQTT → Mosquitto → MQTT exporter → Prometheus → Grafana

- **Custom Python exporter** turns sensor JSON into Prometheus metrics.
- **Alert rules:** sensor offline, exporter down, implausible readings.
- **Grafana dashboard provisioned as code** — no manual clicking.

Results



Live dashboard: temperature, humidity, pressure and Wi-Fi signal for two nodes

Infrastructure as code

- **Docker Compose** — the full backend (broker, exporter, Prometheus, Grafana) runs anywhere with one command.
- **Ansible** — idempotent provisioning of both Raspberry Pis.
- **GitHub Actions CI** — lints configs, validates the stack and **compiles the ESP32 firmware** on every push.

Tech stack

Layer	Technology
Embedded	ESP32, BME280, Arduino/C++
Networking	hostapd, wpa_supplicant, dnsmasq, iptables/NAT
Messaging	MQTT (Mosquitto)
Monitoring	Prometheus, Grafana
Automation	Docker, Ansible, GitHub Actions

Summary

- A complete, **full-stack IoT pipeline** — from sensor firmware to dashboards.
- Runs on **real hardware** *and* as a **containerised** backend.
- **Reproducible, documented and security-aware** (auth, no secrets in git).

Thank you.